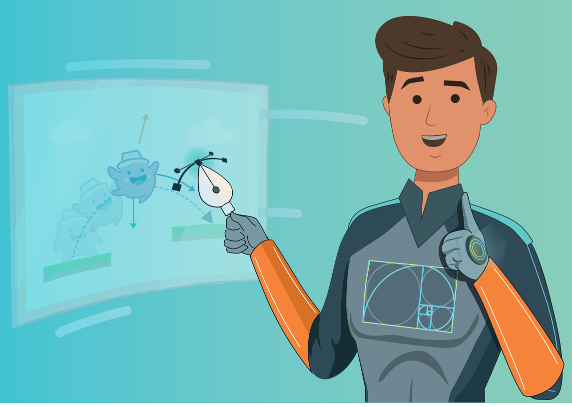


Platformer Movement

10

A game is an activity or sport played competitively or for fun. Some involve physical activity, like cricket, football, etc. Some are played virtually on digital devices, like Candy Crush, Mario Kart, etc. Each game has its own set of rules which players must follow to complete it and win.

10. Platformer Movement

Recall:

A video game is an electronic game where a player interacts with objects and characters displayed on a screen.

A video game involves interaction with a user interface or input device, such as a joystick, controller, keyboard, or motion sensing devices, to generate visual feedback. The feedback can be displayed on a two or three-dimensional video display device such as a TV set, monitor, touchscreen, or virtual reality headset.

What is game design?

Recall:

Game design is the art of using design and aesthetics to create a game.

There are many different types of video games. These are usually categorised by their game characteristics, which refers to how the player interacts with the game.

Types of video games

Game design process

To design a game, we need to follow a specific process. The game design and development process involves the following stages:

Concept → Design → Test → Publish

a. Concept: A concept is the idea for the game. It includes the story, setting, and characters in the game.

b. Design: The design stage involves using a game engine to build the asset library and add functionality.

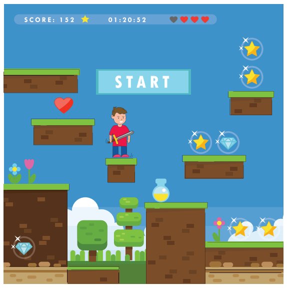
c. Test: In the test stage, the game is tried out on the desired output device and target audience.

d. Publish: If any bugs are found during the test stage, they are resolved, and an updated version of the game is published for the audience.

Platformer Games

A platformer game is one in which a character jumps over obstacles or platforms to reach its goal. The player controls these jumps to make sure their character does not fall or miss a jump. This kind of gameplay is called platforming.

A platformer game involves elements like gravity, wind or other forces as important features of the game.



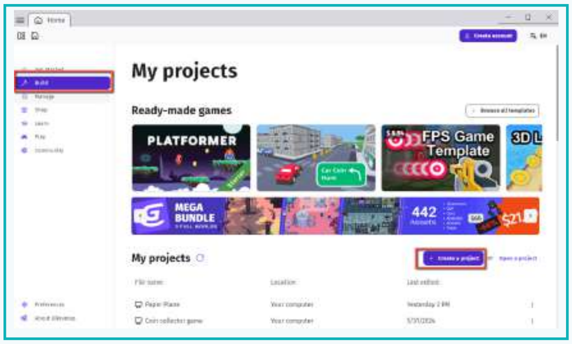
Activity

Understand and create player movement for a platformer game.

A platformer character can move over platforms in either direction and whenever it is grounded or on a platform, it can jump when instructed by the player.

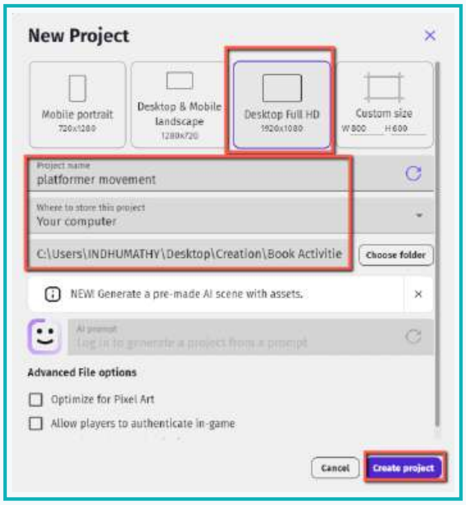
1 Create a new project

a. To start creating a game, open the GDevelop application and create a project.



Activity

b. A new project window will open. Select the 'Desktop Full HD' mode, name the project 'Platformer Movement', save the file in a proper location, and click the 'Create project' button.

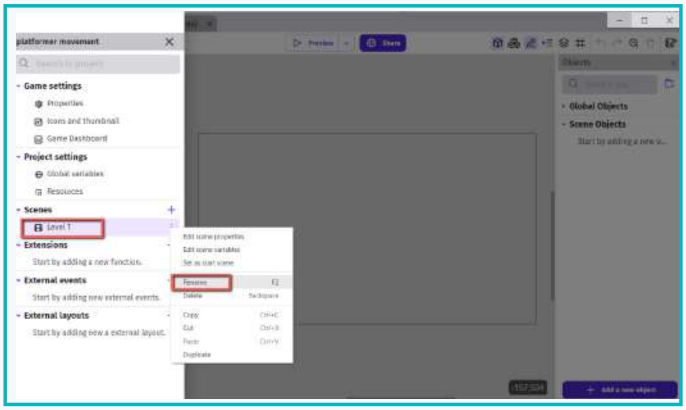


We need an area to play the game in. The area on the screen where the play area is designed is called a scene.

SCENE: The scene is where game levels, menus, inventories, and other game screens are designed and built.

c. To access the default untitled scene, click the 'Project Manager' icon.

d. This creates a new scene. Scenes and assets must be named as they are created.



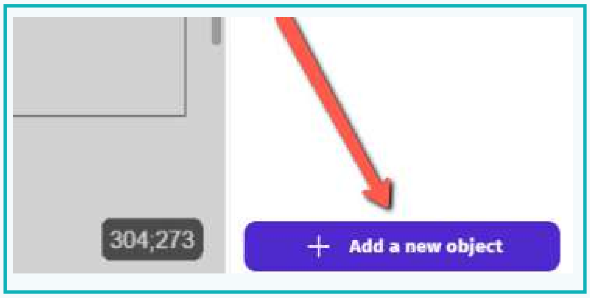
e. Double click on the Level 1 scene to view it. This is where we will design the game levels.

2 Add elements to the scene

OBJECTS: Any visible/invisible element in a game, such as characters, obstacles, enemies, etc. is termed an object.

Here, we will create a character for the game.

- To create an object, click on 'Add a new object' in the Objects panel.

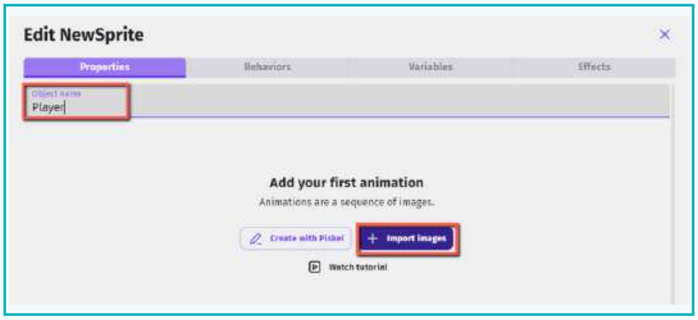


Activity

3 Create a sprite object

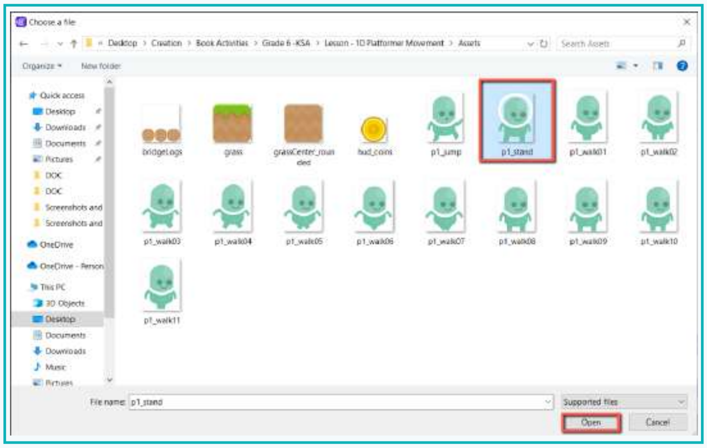
SPRITE: A sprite is a 2-dimensional image. It is a visual representation of an object. It defines the way an object looks.

a. Name the object and click on the 'Import Images' button.



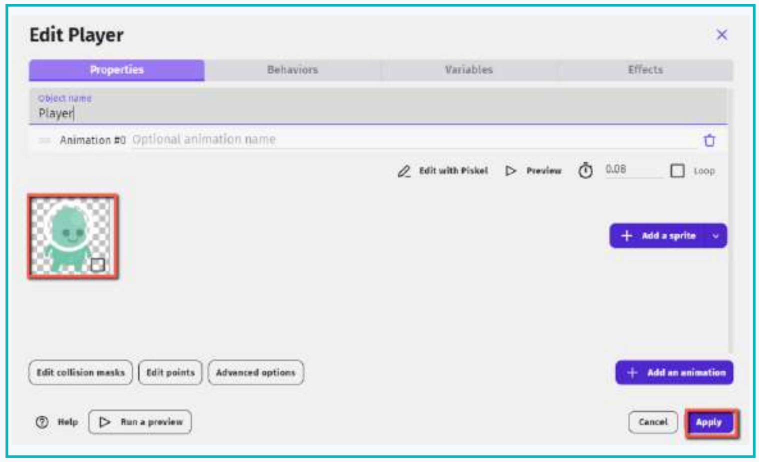
b. Select the character image from the asset folder and click the 'Open' button.

Scan the QR code to download the assets.

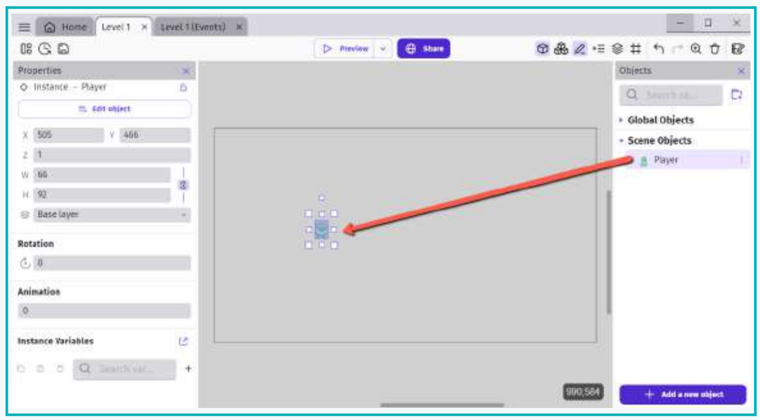


Activity

c. Click the ‘Apply’ button to implement the changes.



d. Select and drag the character object to the scene area.



e. Instruct the object to behave like a platformer character.

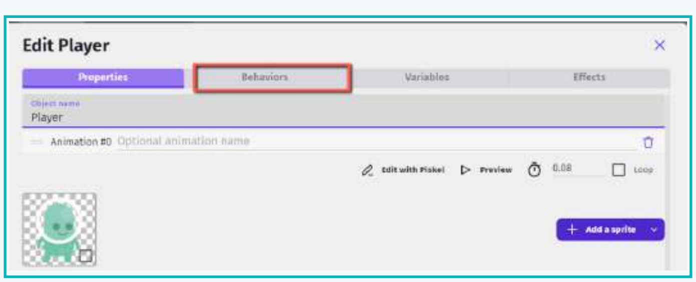
A platformer character object should be able to rest on any platform. When on a platform, it should move forward in response to input keys. Also, while on a platform, the character should be capable of jumping, walking, or running.

4 GDevelop offers platform functionality in the form of behaviors.

BEHAVIOR is a predefined set of actions which adds extra capabilities to the object.

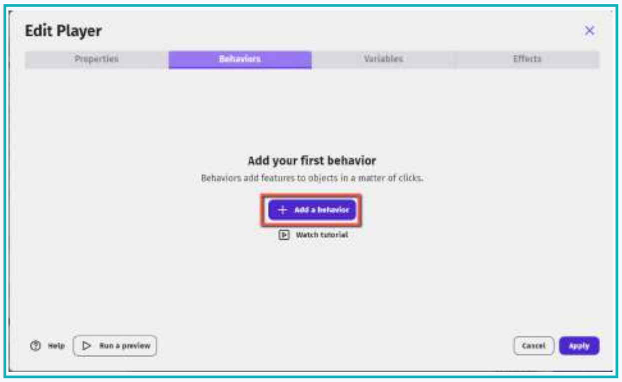
For this game, we will apply ‘Platformer Character’ behavior to the object.

a. Double click on the player object and switch to the behaviors tab.

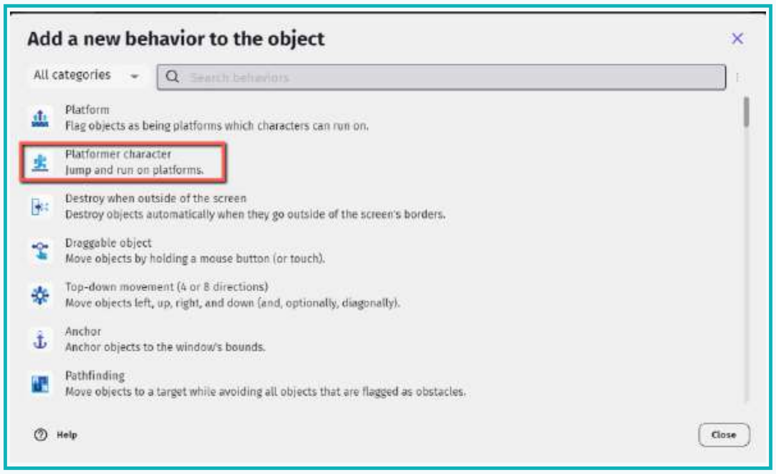


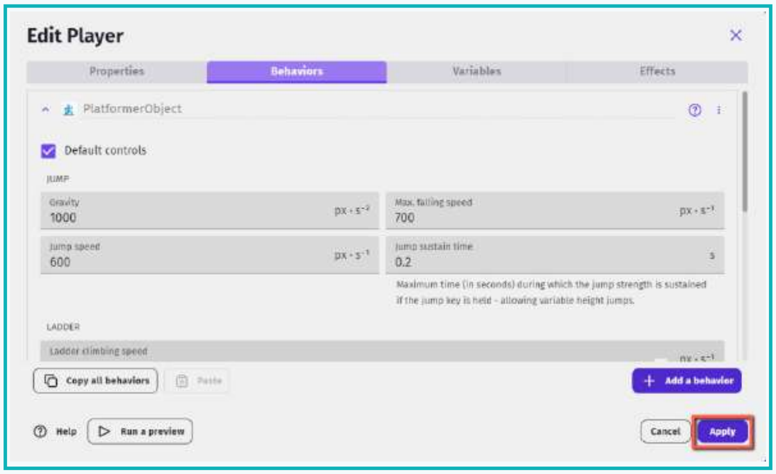
Activity

b. Click ‘Add a behavior’ to the object.

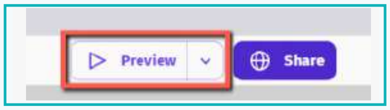


c. Choose the ‘Platformer Character’ behavior. To accept the platformer character’s predefined values, select the ‘Apply’ button.



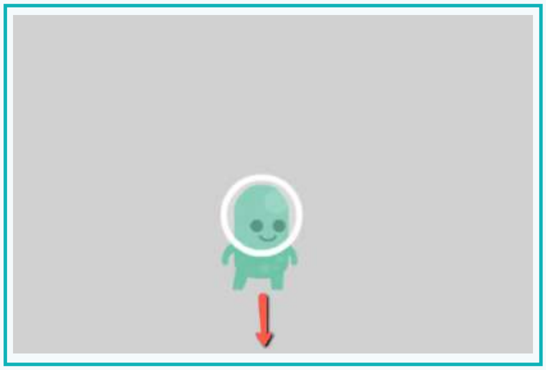


d. Click the ‘Preview’ button in the top menu.

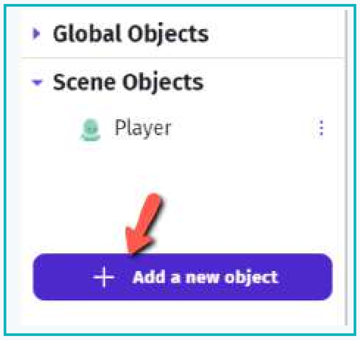


Activity

You should be able to observe the character object falling.



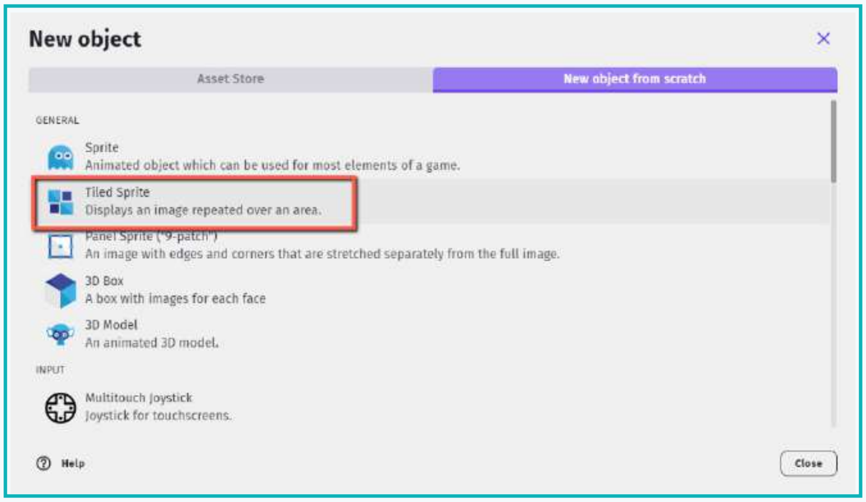
5 Create a new object for the platform (ground) so that the character does not leave the game area. The character should be able to rest and execute movements on the platform.



Different games require different types of platforms. They could be short or long. But the pattern in which the object moves should be consistent. This time, instead of a sprite object we will create a tiled sprite object.

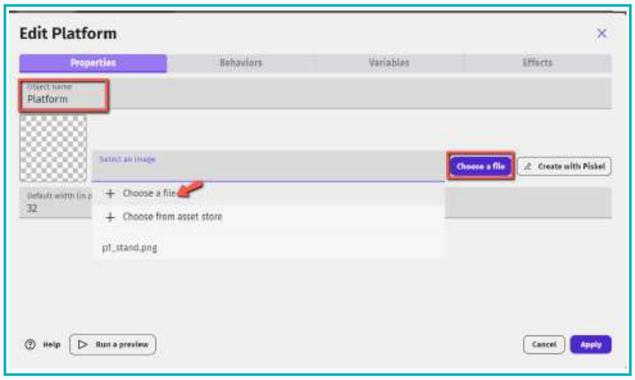
A tiled sprite object allows us to display an image in a tile set. The size of the tile set can be changed by scaling the object in the scene.

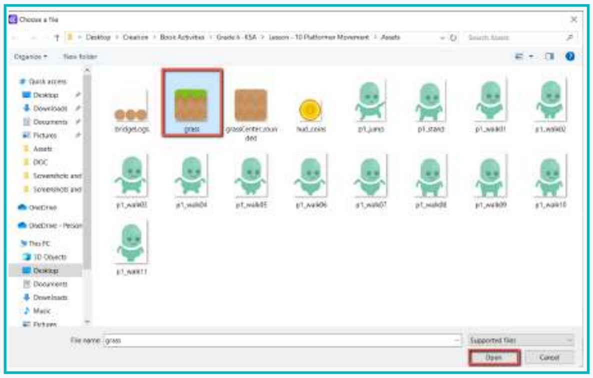
a. Select the object type as tiled sprite.



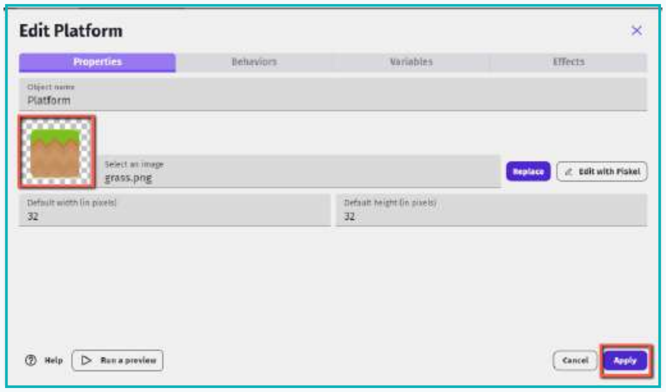
Activity

b. Name the object as 'Platform' and select an image for the ground.

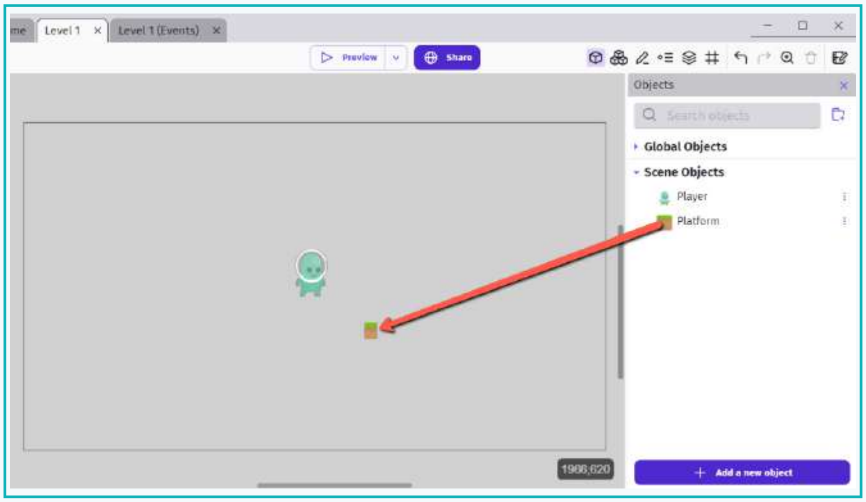




- Once the image is added, click the 'Apply' button.



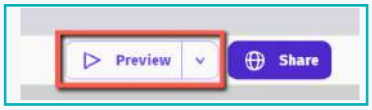
c. Drag the object into the scene area.

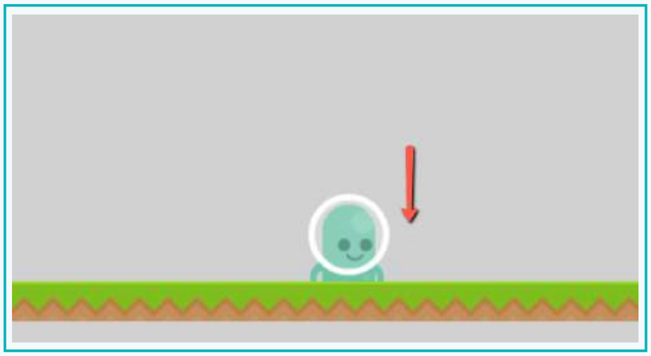


Activity

d. Scale it to create a platform of the desired size.

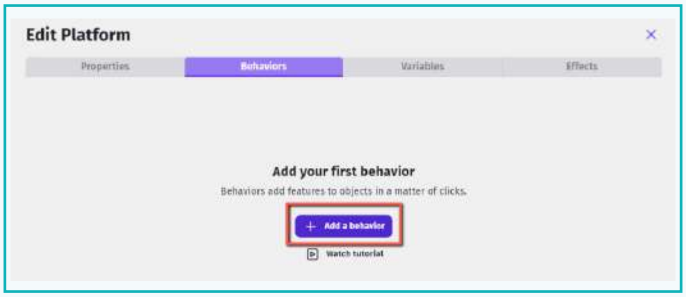
6 Preview the platformer movement game





Observe that the player falls despite the platform being added. To rectify this, we need to instruct the ground object to behave like a platform.

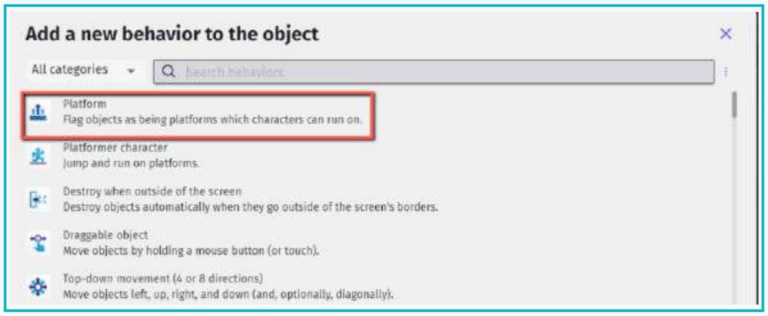
a. Select the platform object and switch to the behaviors tab and click on 'Add a behavior' to the object.

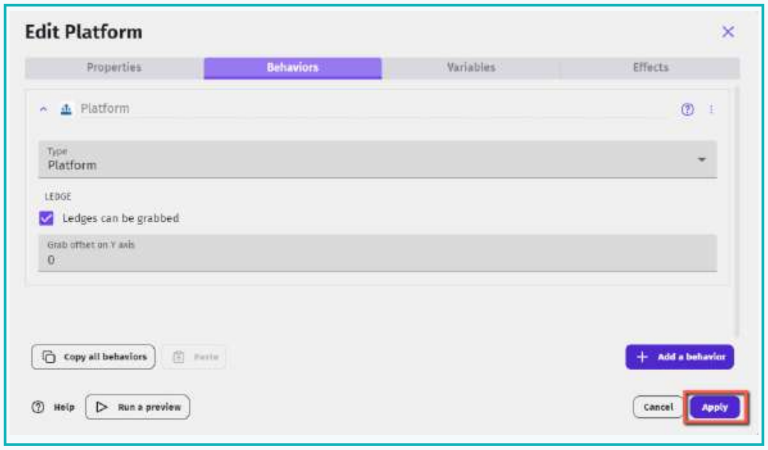


Activity

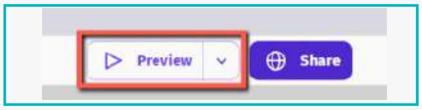
b. Choose the ‘Platform’ as the behavior.

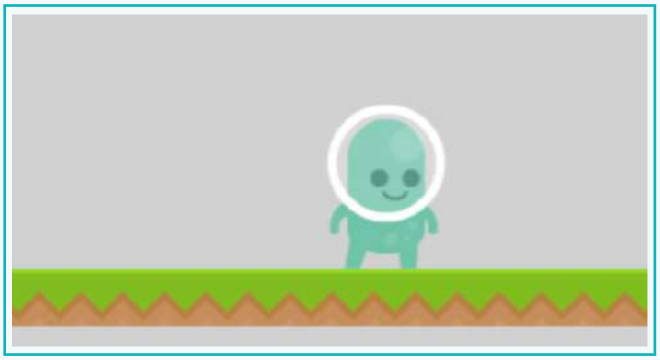
Platform behavior sets the object as a traditional platform. The character can collide with the platform as well as walk on it.





7 Test the game.





Exercise

Answer in one line

1. What are the stages of game design?

▲ ____

____

2. What are behaviors?

▲ ____

____

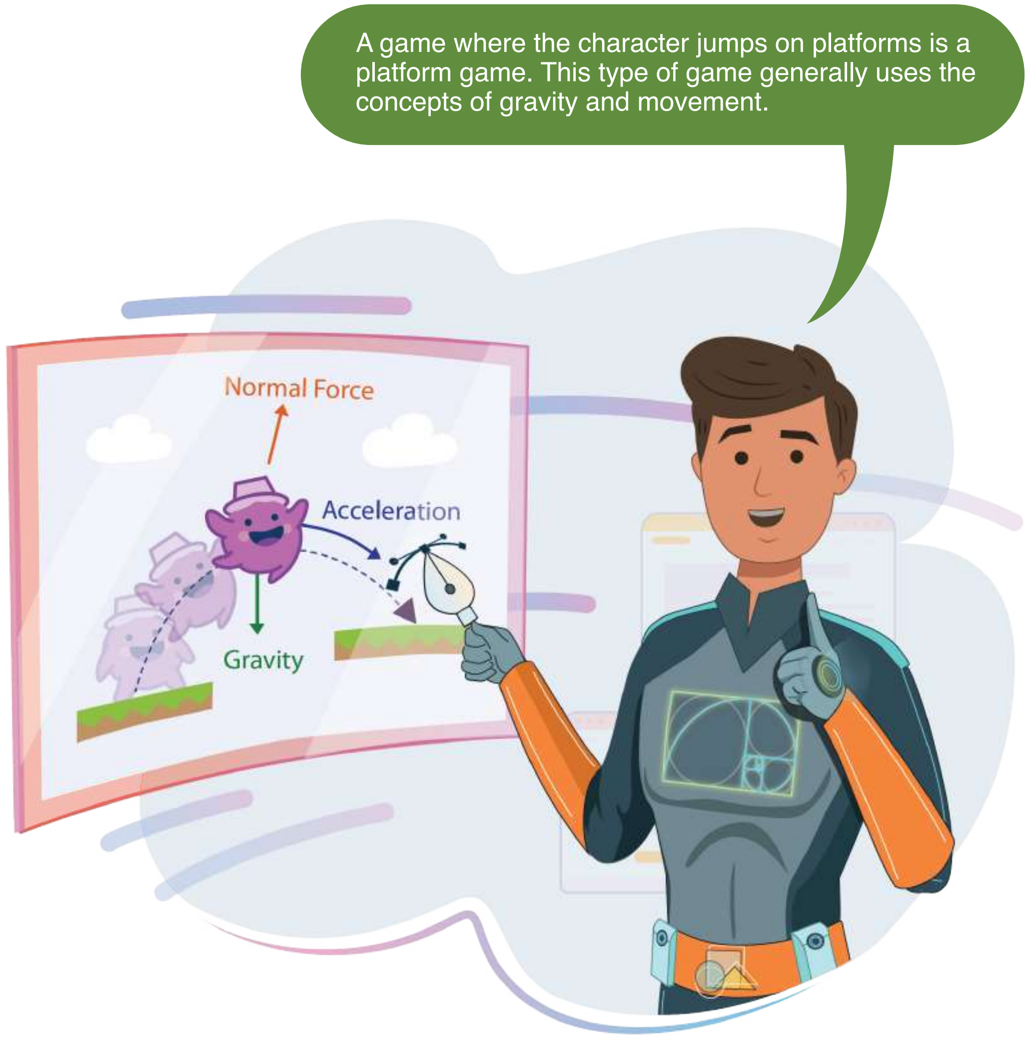
3. What is a platformer game?

▲ ____

____

In this lesson, we learnt how simple and interesting it is to create a platformer game. In this kind of game, the character objects are set on a traditional platform. The character object must cross the platform to reach its goal. Developing our own games is a fascinating process with endless possibilities.

10. Platformer Movement



Tech Flash

- **Sprites** : A sprite is a 2-dimensional image used to visually represent an object.
- **Behaviors** : A behavior is a predefined set of actions that adds extra capabilities to the object.
- **A tiled sprite object** : A tiled sprite allows us to display an image in a tileset.
- **Platform behavior** : Platform behavior sets an object as a traditional platform. The character can collide with the platform and/or walk on it.